STEM Center USA

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S T E M

Analysis Presentation

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01 Introduction

This analysis evaluates STEM Center USA's program data (2020–2024) to measure impact and effectiveness. Using pre- and post-assessment datasets (3,804 vs 3,196 observations post-cleaning and 2,593 vs 2,265 observations post-cleaning, respectively), key variables studied include:

- **Assessment scores:** Comparison of pre-and post-assessment results to evaluate learning gains
- **Program impact:** Influence on STEM interest, coding interest, and college aspirations
- **Engagement metrics:** Program enjoyment, likelihood of recommending, and re-enrollment interest
- **Demographics:** Grade level and program type participation

Data was organized, cleaned, and analyzed in R, with statistical tests such as polychoric correlation and Cramer's V. Visuals and tables highlight trends and insights, providing a clear view of program outcomes, learning gains, and areas for improvement.

02 Strengths & Weaknesses

Strengths

- **Large datasets:** More than 2,000 observations in both pre- and post-assessment datasets for robust analysis.
- **Comprehensive data:** A range of questions allows the exploration of STEM interests, coding, and college aspirations.
- **Longitudinal scope:** Five years of data, 2020–2024, enables trend analysis and tracking progress over time.
- **Quantitative insights:** Statistical tests, including polychoric correlation and Cramer's V, provide quantifiable relationships between variables.
- **Mission alignment:** Analysis aligns with the mission of STEM Center USA to inspire critical thinking and technological literacy.

Weaknesses

- **Lack of Identifiers:** Pre- and post-assessment data could not be linked to individual students.
- **Inconsistencies with the Survey Design:** Variations in survey design impacted uniformity, introduced missing data, and required more cleaning.
- **Qualitative Issues:** Subjective responses, such as names, introduced complexity during data cleaning.
- **Timeline Constraints:** Additional time can help delve into trends and develop extra analyses.
- **Non-random sample:** Data is from students in specific partner schools, limiting generalizability.
- **Grade level imbalance:** Overrepresentation of 4th-8th grade students, with limited data from other grade levels.

03 Methodology

Methodology	Data Type	Key Assumptions	Primary Use Case
Polychoric Correlation	Ordinal	Latent bivariate normal distribution	Correlation analysis for ordinal data
Cramer's V	Nominal	None beyond categorical data	Strength of association in contingency tables
Chi-Square Test	Categorical	Expected cell frequencies ≥ 5	Test of independence or goodness-of-fit
Freeman's Theta	Categorical	None beyond categorical data	Alternative association measure for large tables
F Test for Variance	Continuous	Normal distribution, independent samples	Comparing variances between two populations
Welch Two-Sample t-Test	Continuous	Normal distribution, unequal variances, independent samples	Comparing means with unequal variances

- **Polychoric Correlation**
  - Data Type: Ordinal
  - Purpose: Correlation analysis for ordinal data based on latent normal distribution.
  - Use Case: Survey data with Likert scales.
- **Cramer's V**
  - Data Type: Nominal
  - Purpose: Measures association strength in contingency tables.
  - Use Case: Categorical data relationships.
- **Chi-Square Test**
  - Data Type: Categorical
  - Purpose: Tests independence or goodness-of-fit.
  - Use Case: Frequency analysis with expected cell frequencies ≥ 5.
- **Freeman's Theta**
  - Data Type: Categorical
  - Purpose: Alternative association measures for large tables.
  - Use Case: Nuanced contingency table analysis.
- **F Test for Variance**
  - Data Type: Continuous
  - Purpose: Compares variances of two populations.
  - Use Case: Preliminary variance analysis.
- **Welch Two-Sample t-Test**
  - Data Type: Continuous
  - Purpose: Compares means with unequal variances.
  - Use Case: Robust alternative to standard t-test.

03 Data Breakdown

Figure 1: Counts by Grade

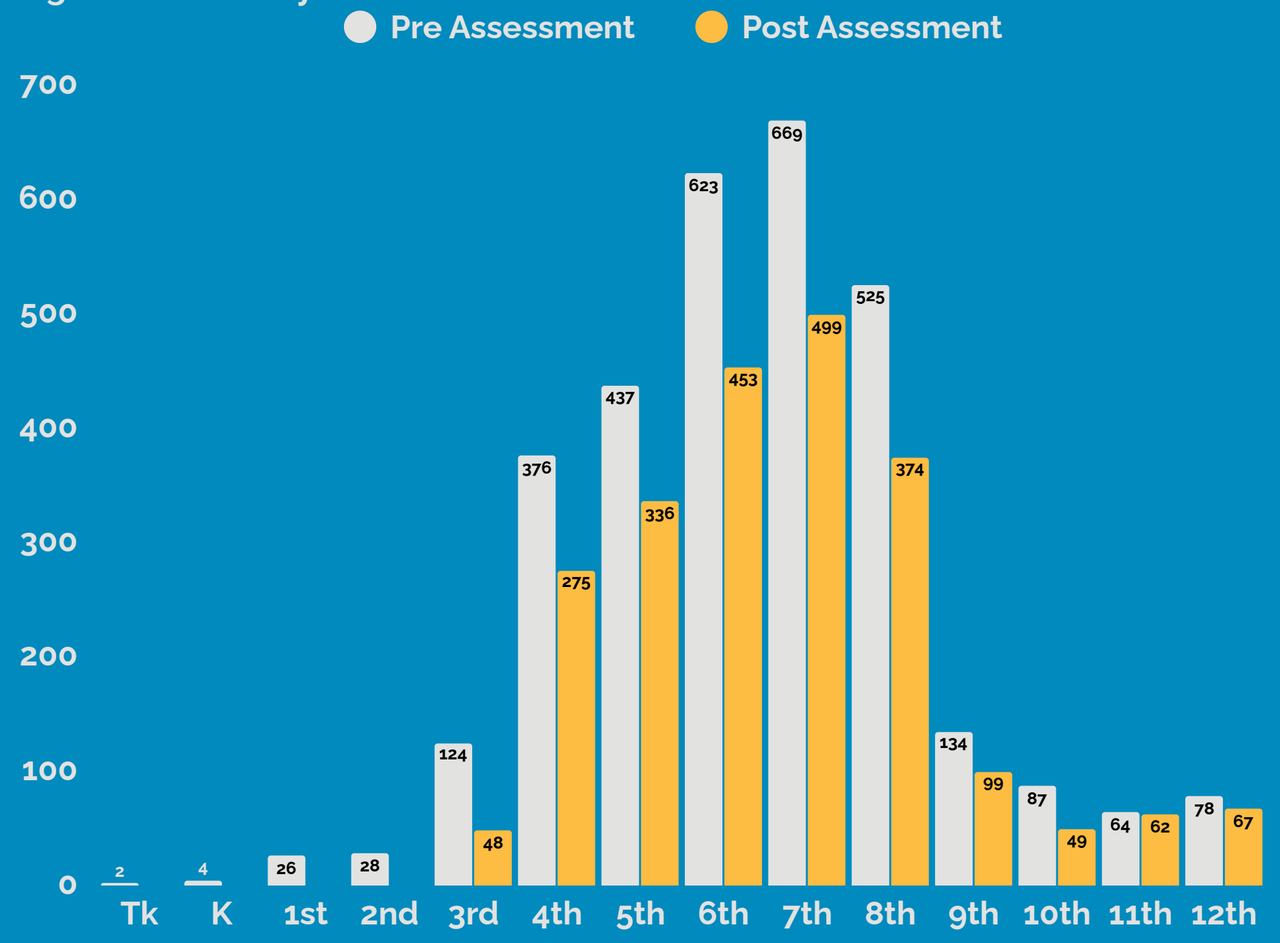
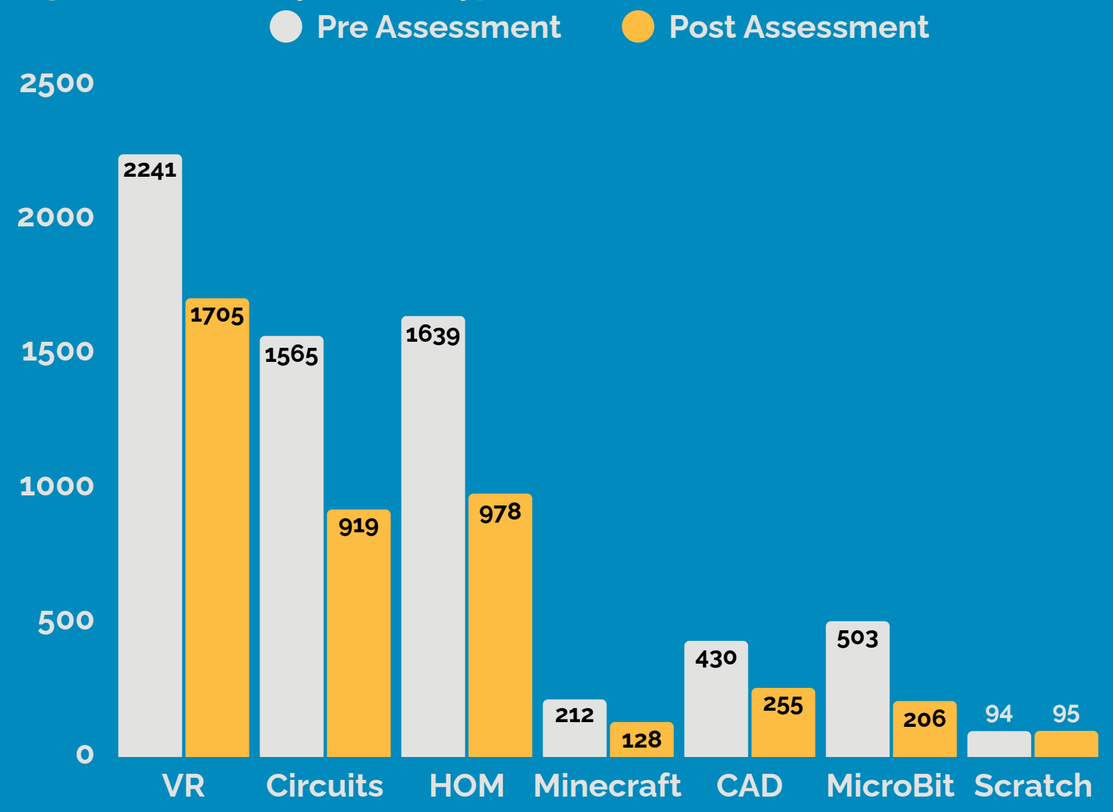


Figure 2: Counts by Course Type



03 Data Breakdown

Figure 3: Counts by Favorite Subject

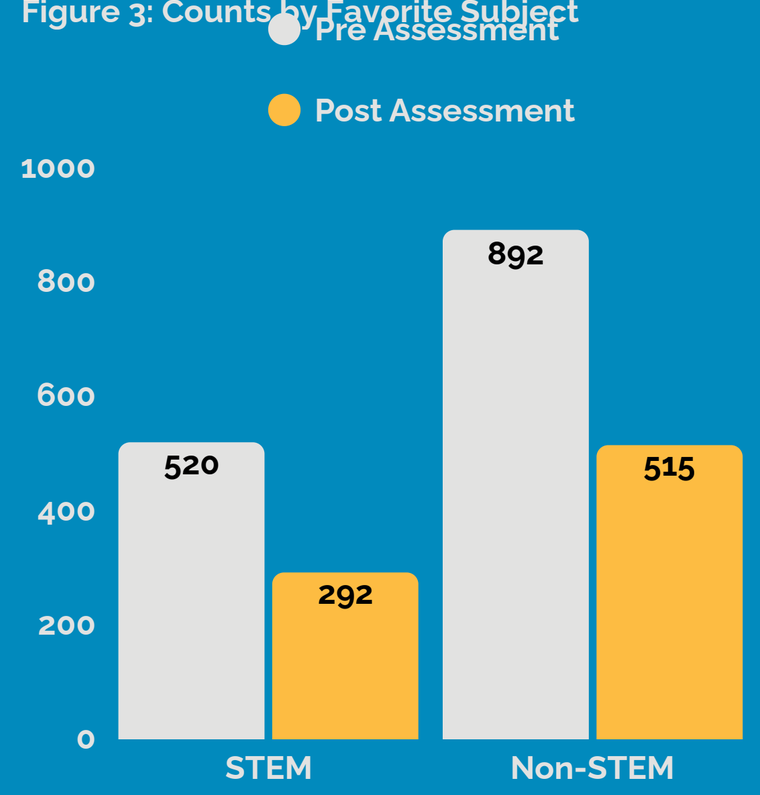


Figure 4: Counts by Year Post Assessment

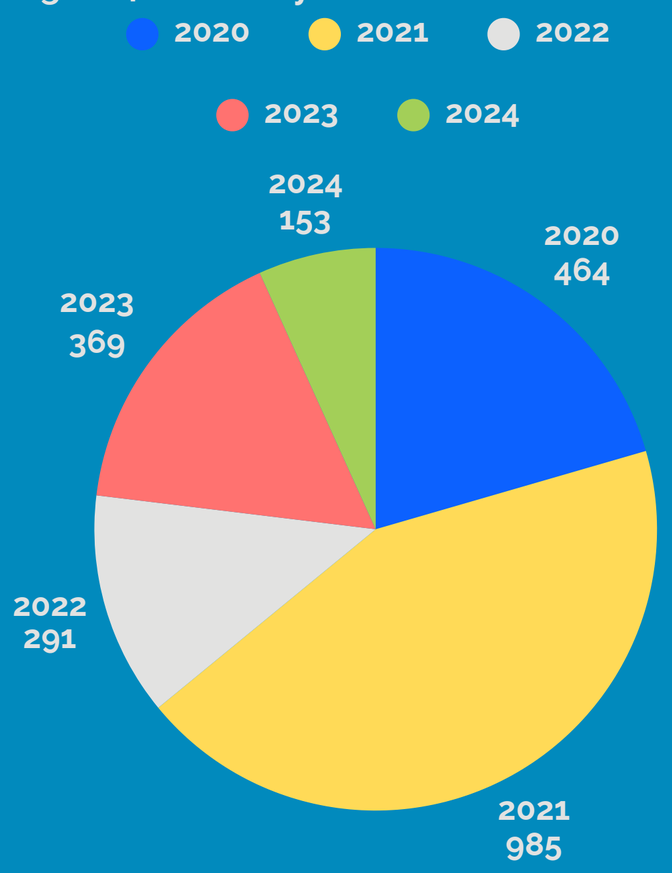
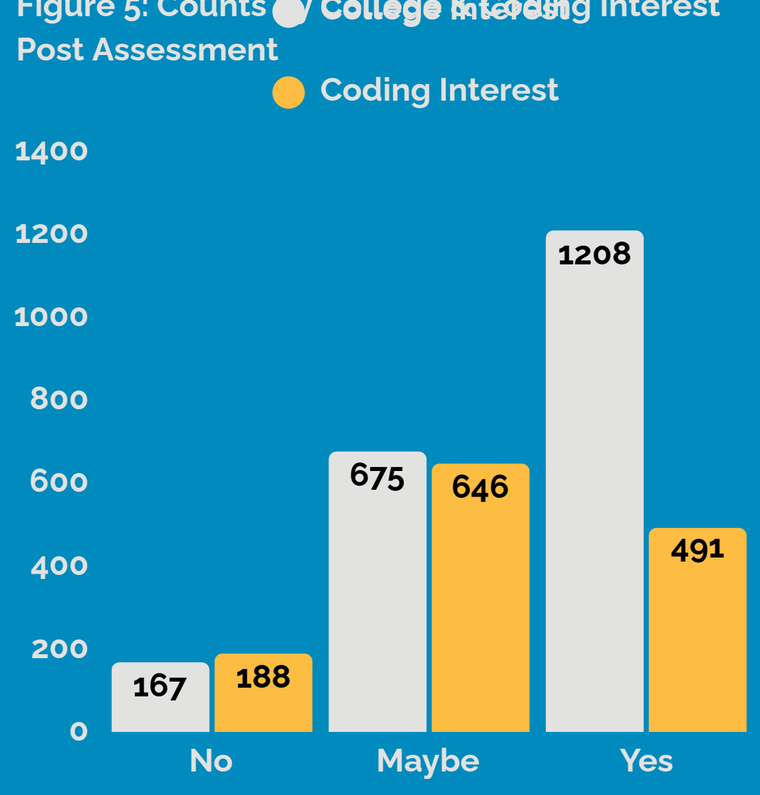
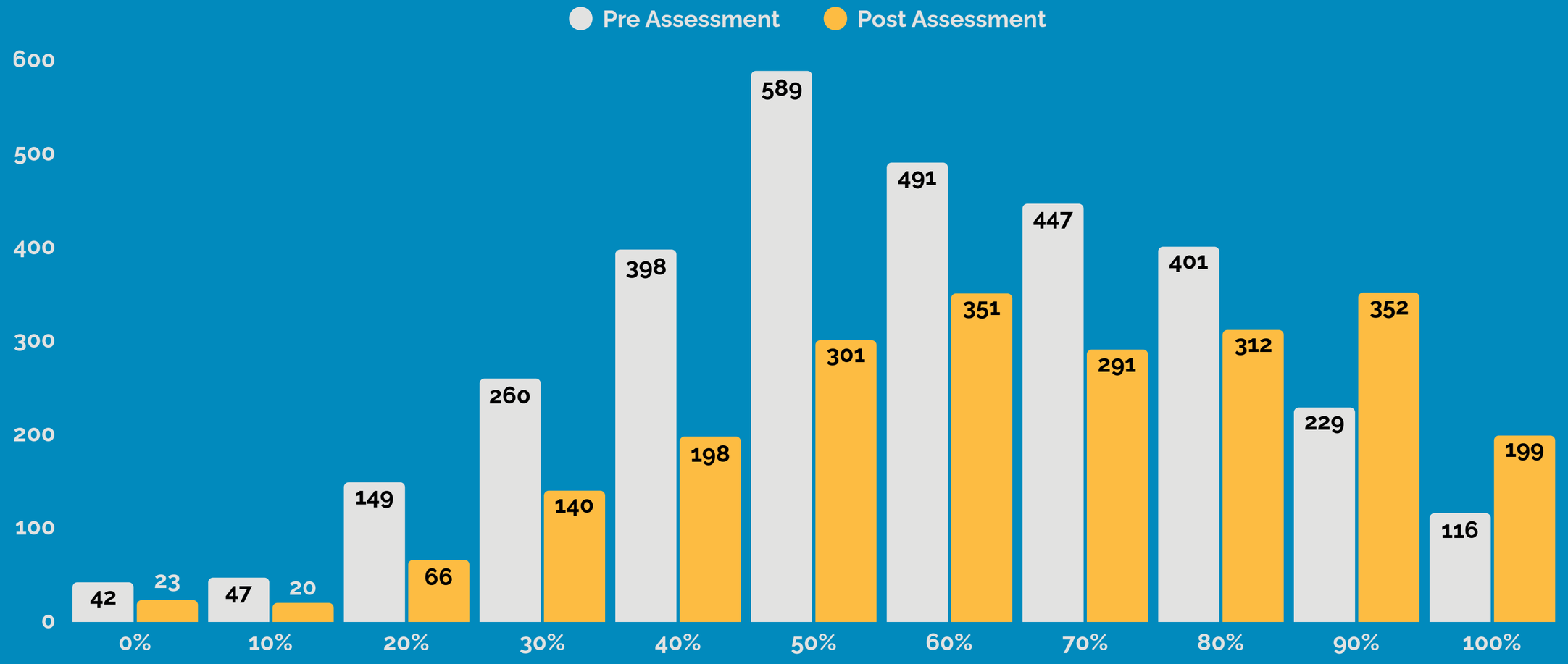


Figure 5: Counts by College & Coding Interest Post Assessment



03 Data Breakdown

Figure 6: Pre vs Post Assessment Score by Percentage



04 Favorite Subject vs College Interest Post

Figure 7:

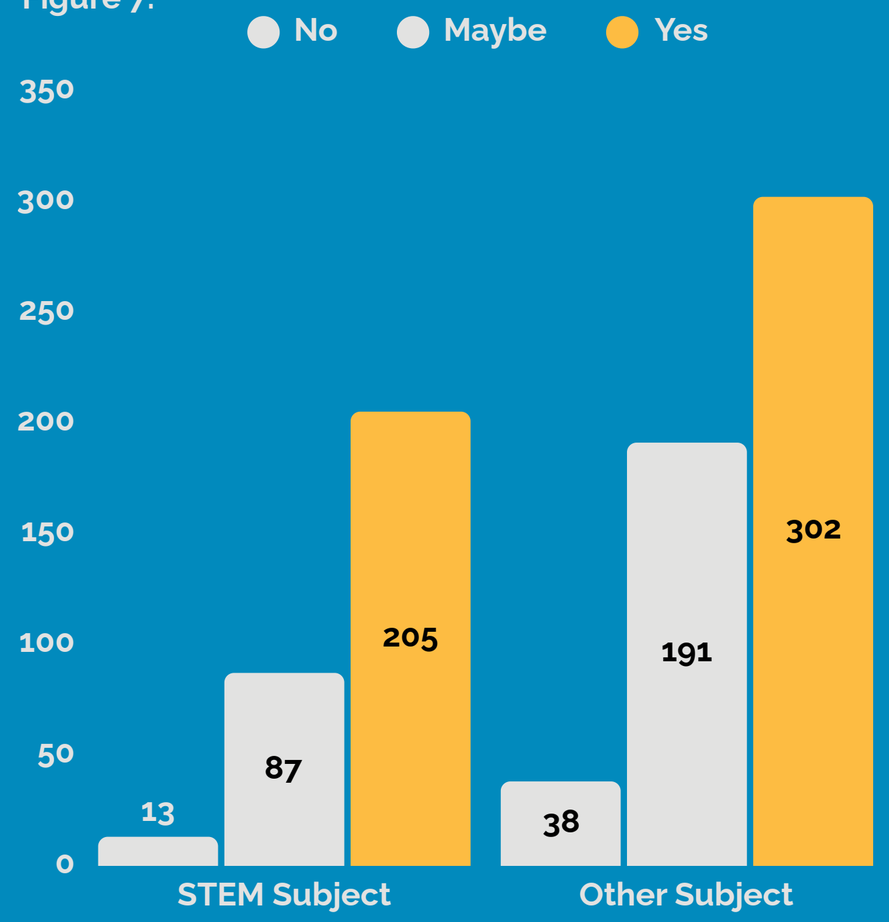


Figure 7 shows counts between students who listed a STEM subject vs any other subject as their favorite and their interest in attending college post-program. N = 836 (only includes students without NA's for both variables)

Results: Looking at counts alone, we see a larger pool of students who chose a non-STEM-related subject as their favorite. However, if we compare their respective proportions, we can see that **those who decided STEM subjects as their favorite were likelier to be interested in college post-program**.

Variable Association: With a chi-square test p-value of 0.009, we have sufficient evidence to say **there is an association between favorite subjects and college interest post-program**. However, with a Cramer's V measure of 0.05, the strength of the relationship is **fairly weak.**

Table 2

College Interest	No	Maybe	Yes
STEM Subject	4%	29%	67%
Other Subject	7%	36%	57%

04 Favorite Subject vs Coding Interest Post

Figure 8:

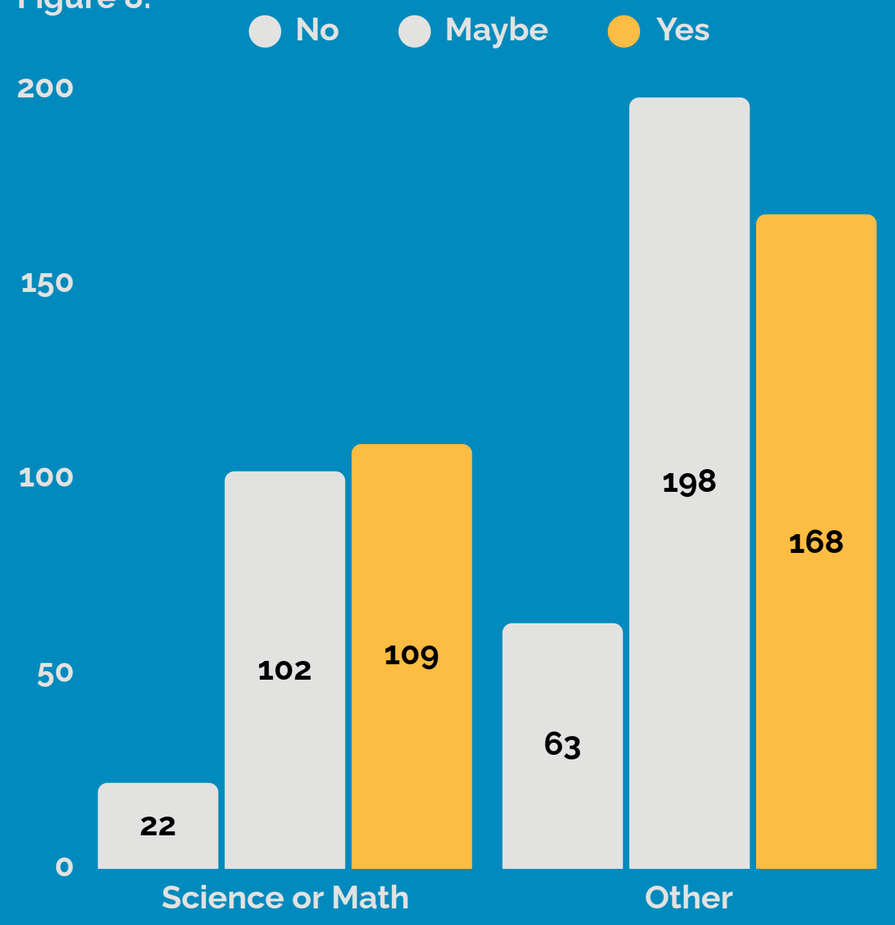


Figure 8 shows counts between students who listed a STEM subject vs any other subject as their favorite and their interest in attending college post-program. N = 662 (only includes students without NA's for both variables)

Results: Similar to Favorite Subject vs College Interest, we see a larger pool of students who chose a non-STEM-related subject as their favorite. However, if we compare their respective proportions, we can see that **those who decided STEM subjects were their favorite, were likelier to be interested in coding post-program.**

Variable Association: A chi-square test resulted in a p-value of 0.063, so we failed to reject the null and **lacked sufficient evidence to say there is an association between favorite subjects and coding interest** post-program. However, there is an apparent difference proportion-wise.

Table 3

Coding Interest	No	Maybe	Yes
STEM Subject	9%	44%	47%
Other Subject	15%	47%	39%

04 Year vs College Interest Post

Figure 9:

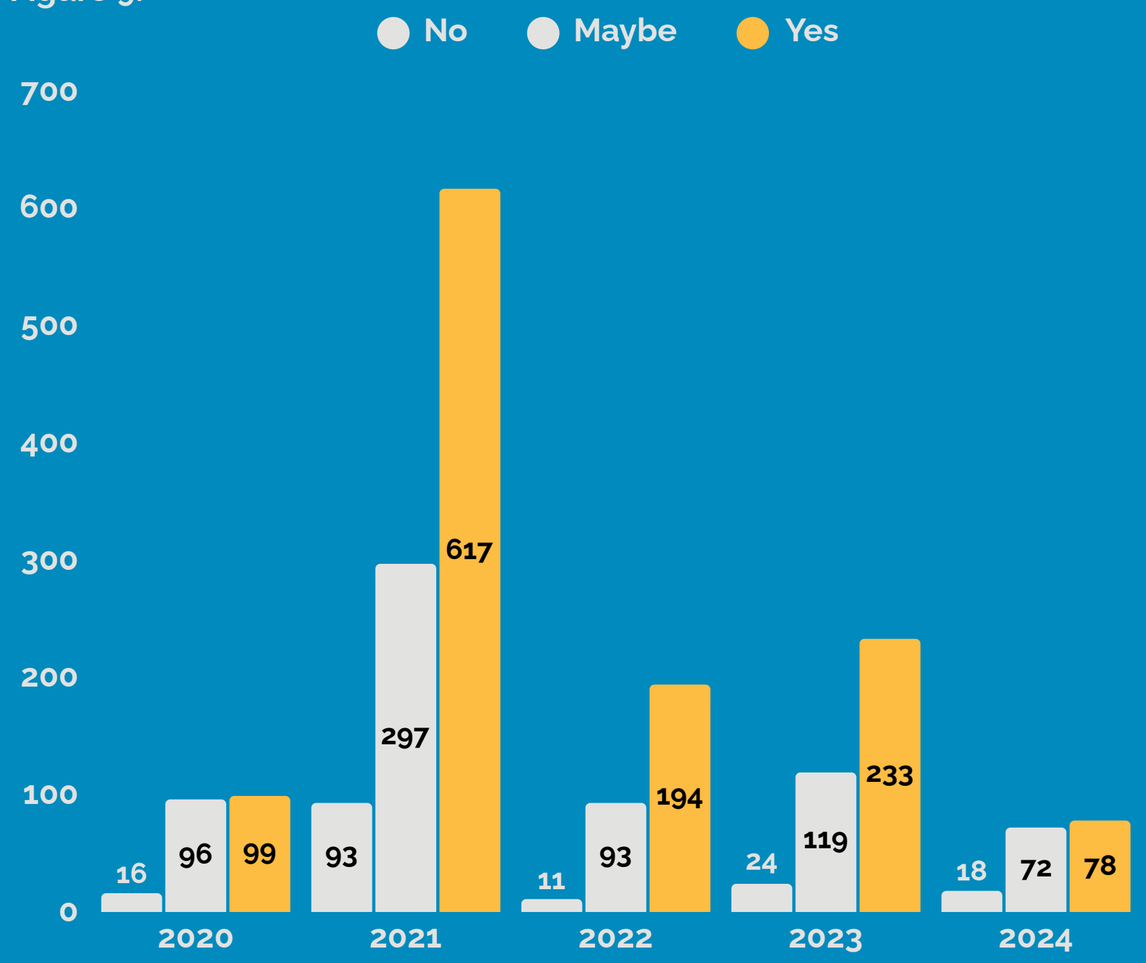


Figure 9 shows counts for years and student interest in attending college post-program. N = 2060 (only includes students without NA's for both variables)

Results: Looking at counts alone, we see a substantially larger pool of students in 2021. If we compare proportions by year, we can see a steady increase in college interest post-program from 2020-2022, but a decline post-2022. However, this may be attributed to the reduced count in 2024 compared to the previous years. The closest in count would be 2020 which showed similar proportions in college interest.

Variable Association: With Freeman's Theta of 0.08, **there was an indication of a very weak association** between the ordinal and nominal variables.

Table 4

College Interest	*No*	*Maybe*	*Yes*
2020 (N = 221)	7%	43%	45%
2021 (N = 1007)	9%	30%	61%
2022 (N = 298)	4%	31%	65%
2023 (N = 376)	6%	32%	62%
2024 (N = 168)	11%	43%	46%

04 Year vs Coding Interest Post

Figure 10:

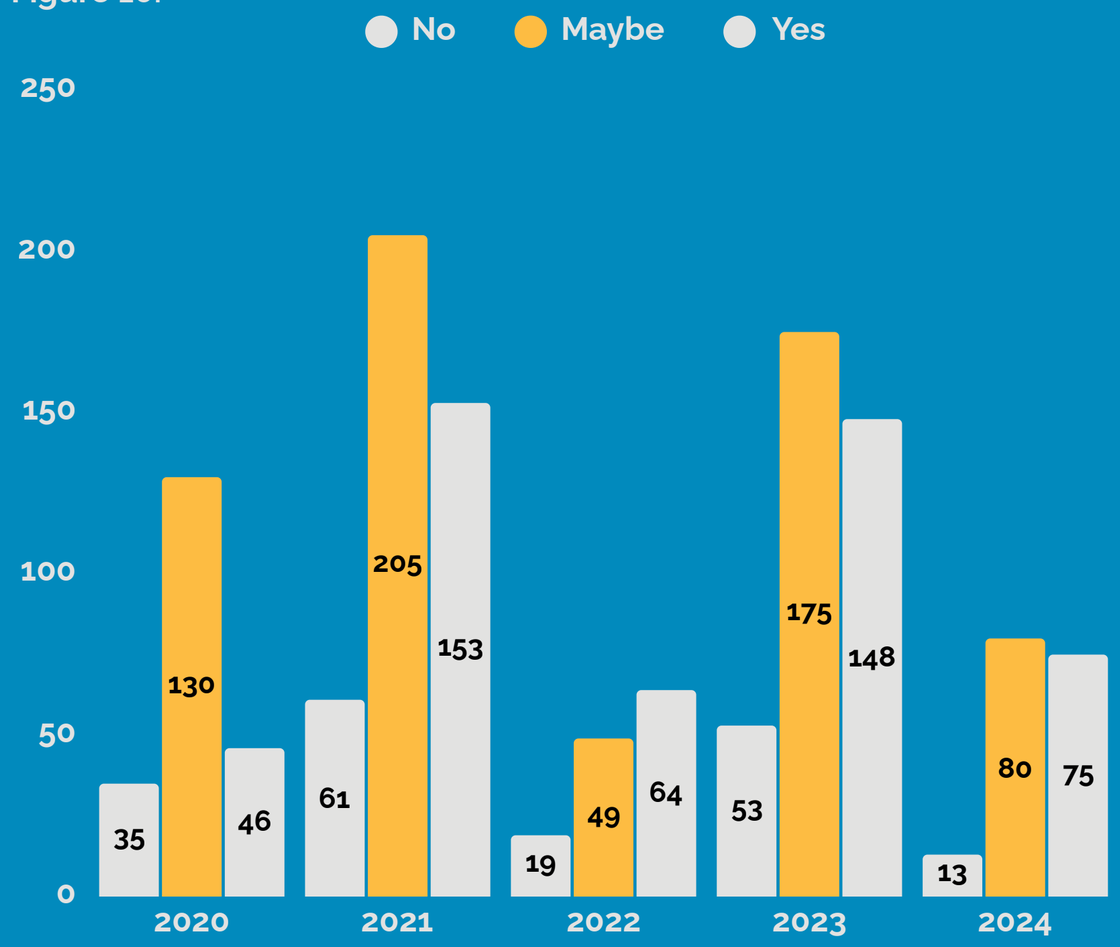


Figure 10 shows counts for years and student interest in coding post-program. N = 1306 (only includes students without NA's for both variables)

Results: Looking at counts alone, we see a substantially larger pool of students in 2021 and 2023. If we compare proportions by year, we can see a steady increase in coding interest post-program from 2020-2022, a decline in 2023, and uptake in 2024. The proportion of students who answered maybe to the impact of the program on their interest in coding is a lot higher than those who answered yes every year except 2022.

Variable Association: With Freeman's Theta of 0.105, **there was an indication of a very weak association** between the ordinal and nominal variables.

Table 5

Coding Interest	No	Maybe	Yes
2020 (N = 211)	17%	62%	22%
2021 (N = 419)	15%	49%	37%
2022 (N = 132)	14%	37%	49%
2023 (N = 376)	14%	47%	39%
2024 (N = 168)	8%	48%	45%

04 Grade vs College Interest Post

Table 6 shows the proportions of grade level and student interest in college post-program. **N = 2060** (only includes students without NA's for both variables)

Results: As previously stated, we overrepresent 4th-8th grade students, so representation in grades outside that range may affect proportions significantly. We can see **steady interest in college post-program across grade levels**, especially those with higher counts (4th-8th). There was a **spike in students** who were **interested in college** for grades **11th and 12th**, and a **dip in** those not interested in college for **grade 3**, but due to the minimal response rate for such years, this may be off from the true proportions.

Variable Association: With Freeman's Theta of **0.0541,** there was an indication of a **very weak or negligible association** between the ordinal and nominal variables.

Table 6

Grade	3rd (47)	4th (264)	5th (333)	6th (412)	7th (405)	8th (343)	9th (77)	10th (50)	11th (63)	12th (66)
No	19%	8%	8%	5%	11%	9%	3%	6%	6%	2%
Maybe	26%	33%	32%	34%	32%	37%	32%	34%	24%	29%
Yes	55%	59%	60%	60%	58%	54%	65%	60%	70%	70%

04 Grade vs Coding Interest Post

Table 7 shows the proportions of grade level and student interest in coding post-program. **N = 1302** (only includes students without NA's for both variables)

Results: Overrepresentation of 4th-8th grade students is shown, and representation outside that range may affect proportions. We can see that **interest in coding post-program**, across grade levels, **is centered more around maybe interested** vs not interested or yes interested, **except for grades 4 and 11**. Again, these discrepancies may be due to the minimal response rate for certain years, although grade 4 is interesting,

Variable Association: With Freeman's Theta of **0.0831**, there was an indication of a **very weak or negligible association** between the ordinal and nominal variables.

Table 7

Grade	3rd (11)	4th (141)	5th (191)	6th (259)	7th (248)	8th (240)	9th (58)	10th (43)	11th (50)	12th (65)
No	0%	14%	14%	13%	17%	15%	10%	12%	16%	6%
Maybe	55%	37%	44%	47%	52%	57%	53%	47%	42%	57%
Yes	45%	49%	42%	39%	31%	28%	36%	42%	42%	37%

05 Avg. Pre vs Post Assessment Scores

Figure 11:

```
        F test to compare two variances

data:  score_per by post_pre
F = 1.1049, num df = 2252, denom df = 3168, p-value = 0.01023
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 1.023871 1.192941
sample estimates:
ratio of variances
          1.104902

        Welch Two Sample t-test

data:  score_per by post_pre
t = 12.443, df = 4699.4, p-value < 2.2e-16
alternative hypothesis: true difference in means between group post and
group pre is not equal to 0
95 percent confidence interval:
 0.06546900 0.08995803
sample estimates:
mean in group post  mean in group pre
         0.6509543         0.5732408
```

Variance is a measure of how far a set of numbers is spread out from their average value. To conduct a T-test, the variance of a variable should be the same across a sample or group of samples.

With a Variance Test p-value of 0.01023 < 0.05, we can't conclude that **there is no significant difference between the two variances, so a Welch Two Sample t-test was performed**.

Overall Change in Learning Outcomes

- Pre-assessment Mean Score: **0.573 (57.3%)**
- Post-assessment Mean Score: **0.651 (65.1%)**
- Mean Improvement: **0.078 (7.8%)**

The data shows a **statistically significant improvement in learning outcomes overall**, as indicated by a Welch Two Sample t-test p-value of 2.2e-16 < 0.05.

05 Avg. Pre vs Post Assessment Scores by Grade

Figure 13:

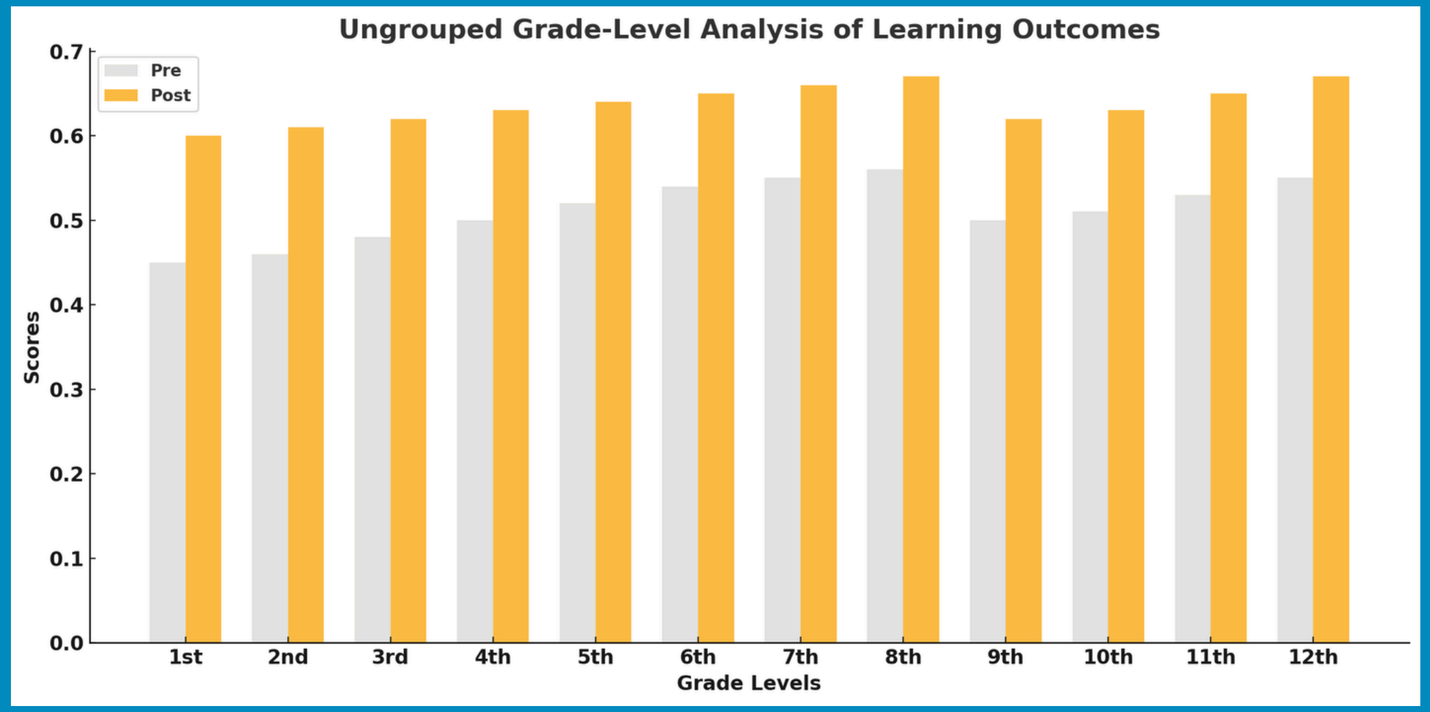


Figure 13 shows a bar chart showing pre and post-mean scores for grade levels (1–12th).

- **Grade-Level Analysis:**
  - Lower grades (1–4): **Lower pre-scores but showed notable improvement.**
  - Middle grades (5–8): **Higher starting scores and consistent improvement.**
  - High school (9–12): Moderate improvement, with higher variability.

05 Avg. Pre vs Post Assessment Scores by Course

Figure 12:

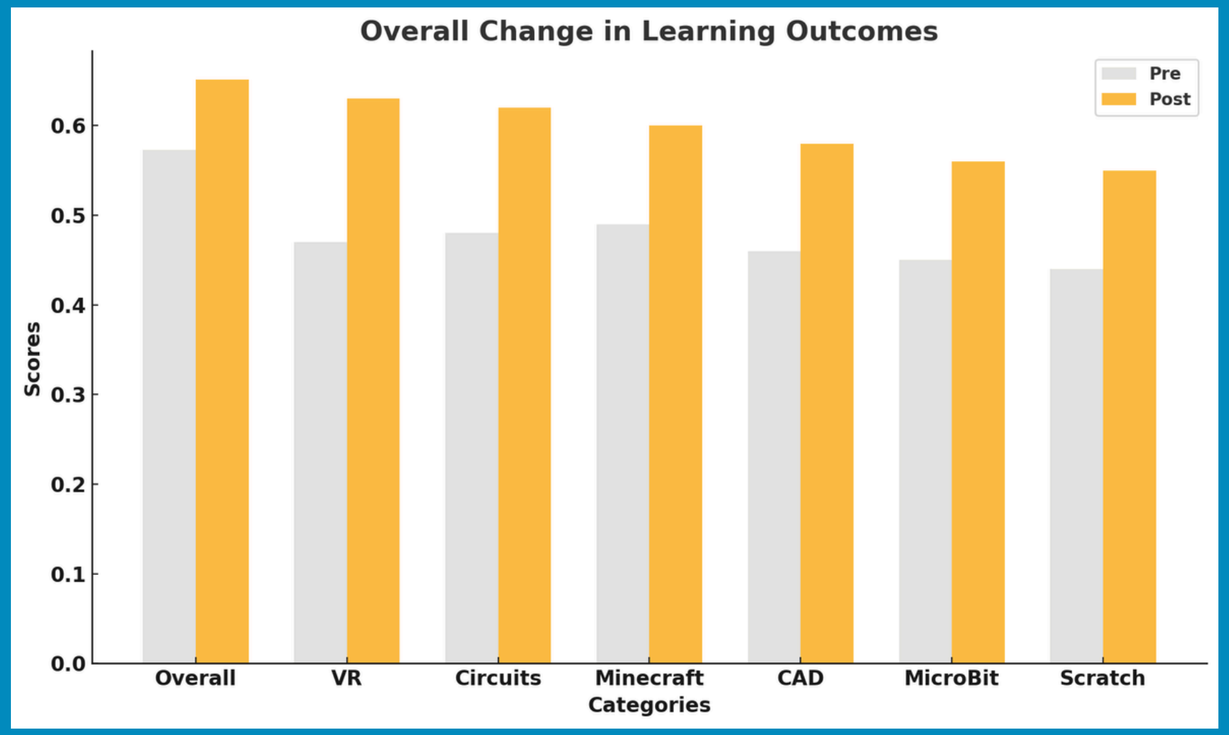


Figure 12 shows a bar chart comparing pre- and post-mean scores across all categories.

Comparison by Course

The percentage improvement in scores across different courses highlights curriculum strengths and weaknesses:

- Virtual Reality:
  - Pre: Concentrated between 0.4–0.6 scores
  - Post: Shifted to **0.6–0.9, suggesting strong improvement**.
- Circuits:
  - Pre: Many scores in 0.3–0.5 range.
  - Post: **Improved distribution in 0.6–0.9 range.**
- Minecraft:
  - Pre: Scores mostly clustered in 0.4–0.5 range.
  - Post: **Significant improvements in higher score ranges (0.7–0.9).**
- CAD:
  - Pre: Scores between 0.4–0.6.
  - Post: **Shift to 0.6–0.8, indicating moderate improvement.**
- MicroBit and Scratch: Showed lesser improvement compared to other courses, with scores remaining relatively clustered in mid-ranges.

05 Avg. Improvement Scores by Grade Level

Table 8												
Grade	VR Pre	VR Post	Circuit Pre	Circuits Post	HOM Pre	HOM Post	Minecraft Pre	Minecraft Post	CAD Pre	CAD Post	Microbit Pre	Microbit Post
3	0.75	0.57	0.71	0.62	0.71	0.62	0	0.4	0	0	0.55	0.4
4	0.63	0.58	0.58	0.61	0.6	0.61	0.47	0.39	0.59	0.7	0.37	0.5
5	0.58	0.61	0.58	0.61	0.58	0.61	0.4	0.47	0.55	0.61	0.51	0.43
6	0.46	0.66	0.42	0.66	0.42	0.66	0.64	0.51	0.59	0.56	0.62	0.51
7	0.51	0.69	0.53	0.6	0.51	0.61	0.59	0.47	0.55	0.55	0.56	0.46
8	0.6	0.72	0.6	0.68	0.61	0.69	0.53	0.53	0.56	0.51	0.5	0.47
9	0.57	0.75	0.61	0.62	0.61	0.62	0.53	0.44	0.5	0.6	0.51	0.48
10	0.63	0.85	0.66	0.65	0.66	0.66	0.61	0.52	0.56	0.59	0.54	0.5
11	0.64	0.79	0.66	0.63	0.67	0.64	0	0.57	0.58	0.56	0.55	0.52
12	0.66	0.68	0.65	0.57	0.66	0.57	0	1	0.51	0.61	0.47	0.51

Table 8: Analysis of Avg. Improvement Scores by Grade Level

1. **General Improvement:** All program types show improvement from pre to post-assessment, with Virtual Reality (VR) and CAD showing the most significant increases across all grades.
2. **Grade-level Trends:** Younger students (3rd-5th grade) show more pronounced increases in mean scores across the board. This suggests that early interventions or learning methods, particularly with hands-on and interactive programs like VR and Circuits, may be more effective.
3. **Older Students:** The improvement for older students (grades 9–12) is still significant, but the growth is generally smaller compared to younger students. This could be due to a saturation effect or that older students may be more experienced and require more advanced content to see significant gains.
4. **Program-Specific Patterns:** VR and CAD seem to have the highest increases, suggesting they may be more engaging or effective in improving skills. Minecraft shows modest improvements, indicating that it may require further content adjustment to maximize learning.

05 Avg. Improvement Scores by Grade Level

Table 9												
Grade	VR Pre	VR Post	Circuit Pre	Circuits Post	HOM Pre	HOM Post	Minecraft Pre	Minecraft Post	CAD Pre	CAD Post	Microbit Pre	Microbit Post
3	55	44	48	31	48	31	0	1	0	0	19	4
4	297	248	230	142	240	137	8	4	3	1	81	31
5	341	280	282	170	295	166	13	9	30	8	99	45
6	439	353	282	178	301	204	23	19	99	44	98	35
7	513	358	267	158	282	183	81	55	80	41	91	29
8	366	255	220	123	237	140	54	32	92	46	55	17
9	78	76	74	30	74	29	21	3	27	19	7	5
10	55	29	50	26	50	27	12	4	20	16	3	3
11	31	34	43	29	43	29	0	1	38	39	15	16
12	47	31	50	35	50	35	0	0	41	41	16	21

Table 9: Analysis of Avg. Improvement Scores by Grade Level

- The counts by grade for the pre- and post-assessment data are crucial. Programs with smaller counts may not show as accurate a reflection of performance, and this can skew the analysis.
- To properly account for the counts by grade for each program type in the analysis, they should be incorporated by weighting the mean scores based on these counts.
- The weighted mean scores by grade are the number of students in each grade and program type.
- This ensures that smaller sample sizes are appropriately reflected in the analysis to ensure the results are more accurate and meaningful.

05 Weighted Avg. Improvement Scores by Grade

Table 10												
Grade	VR Pre	VR Post	Circuit Pre	Circuits Post	HOM Pre	HOM Post	Minecraft Pre	Minecraft Post	CAD Pre	CAD Post	Microbit Pre	Microbit Post
3	0.8	0.55	0.8	0.64	0.8	0.64	0	0.41	0	0	0.55	0.4
4	0.71	0.56	0.61	0.62	0.61	0.62	0.48	0.39	0.62	0.68	0.38	0.48
5	0.57	0.6	0.58	0.62	0.58	0.62	0.47	0.48	0.59	0.62	0.57	0.44
6	0.42	0.7	0.39	0.68	0.39	0.68	0.67	0.49	0.63	0.51	0.65	0.54
7	0.51	0.73	0.52	0.58	0.52	0.59	0.59	0.48	0.6	0.57	0.58	0.43
8	0.6	0.79	0.62	0.73	0.62	0.74	0.54	0.53	0.62	0.55	0.48	0.48
9	0.56	0.84	0.64	0.61	0.64	0.61	0.51	0.41	0.53	0.61	0.48	0.49
10	0.63	0.89	0.71	0.64	0.71	0.65	0.64	0.51	0.61	0.62	0.57	0.49
11	0.67	0.83	0.76	0.61	0.76	0.62	0	0.51	0.68	0.49	0.54	0.49
12	0.65	0.72	0.69	0.59	0.69	0.59	0	1	0.58	0.69	0.46	0.49

Table 10: Weighted Analysis of Avg. Improvement Scores by Grade Level

1. **Virtual Reality (VR):** Post-assessment scores are generally higher than pre-assessment scores. The scores for VR improve significantly in grades 6 and higher, with a noticeable increase in post-assessment performance.
2. **Circuits:** The weighted mean scores show moderate improvements. Circuits have solid performance in post-assessment across grades 6–12, with some increase in performance.
3. **Hands-on Modules (HOM): This shows** a similar trend, with post-assessment scores increasing but not as significantly as in VR. The lower grades, particularly 3rd and 4th, show higher pre-scores, with post-assessment scores stabilizing.
4. **Minecraft:** The improvements are subtle, with post-assessment scores remaining relatively close to pre-assessment scores, showing minimal growth.
5. **CAD:** Shows significant improvements, especially in grades 7–12. The post-assessment performance is better than the pre-assessment across all grades.
6. **Microbit:** The trend shows increases in post-assessment scores, especially in grades 6–9, but it does not perform as strongly as the other programs.

05

Avg. Pre vs Post Assessment Scores by Favorite Subject

Figure 14:

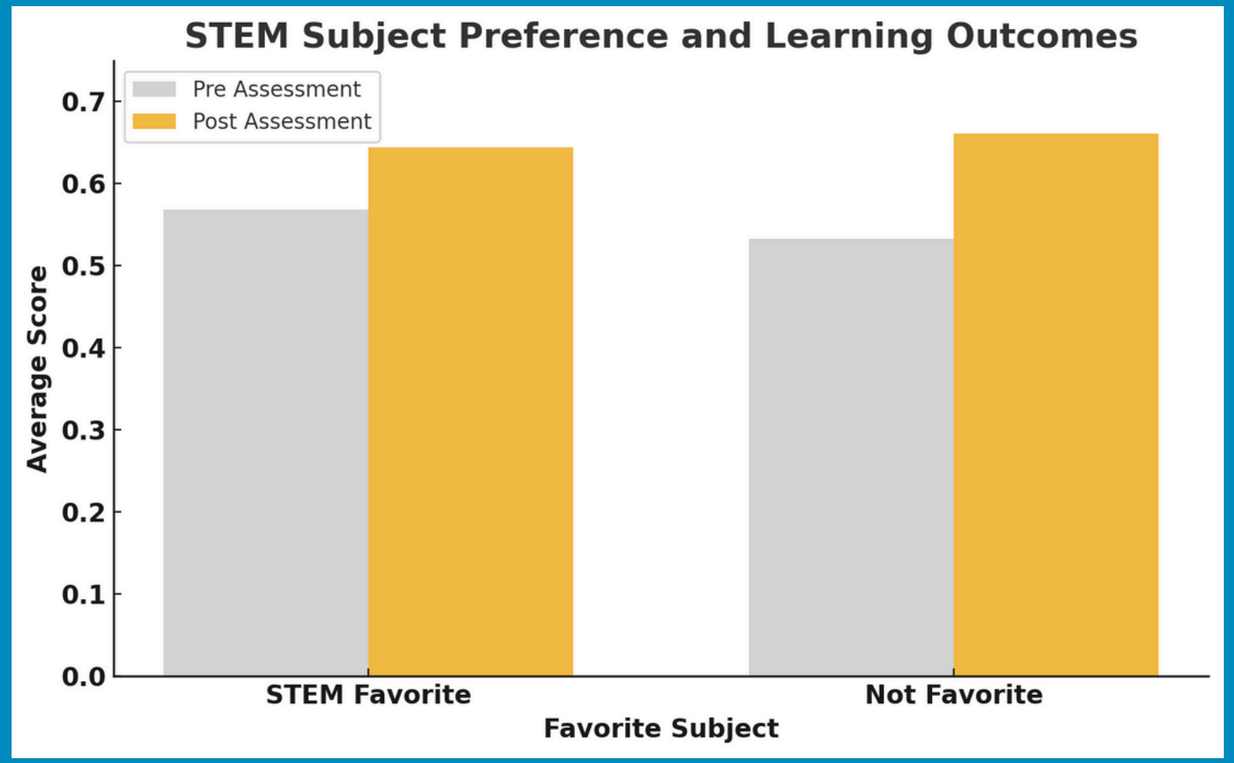


Figure 14 shows a bar chart showing pre and post-mean scores for STEM vs Non-STEM Favorite Subject.

- **Favorite Subject:**
  - In the **pre-assessment,** students who preferred **STEM scored higher than those who didn't**.
  - In the **post-assessment**, the **performance flipped slightly**, with non-STEM-preference students scoring just a bit higher.
  - Overall, **both groups improved**, but the gap **narrowed** and **reversed**, **suggesting more equal outcomes after the program.**

05

Avg. Pre vs Post Assessment Scores by Favorite Subject

Chi-Squared Test Results: STEM Preference vs. Assessment Scores

- Pre-Assessment
  - X^2 = 16.426, df = 10, **p = 0.088**
  - Interpretation:
    - This p-value is above 0.05, so we fail to reject the null hypothesis.
    - However, it's **borderline, suggesting there may be a weak or emerging association between liking STEM and score distribution before the program**.
    - Combined with the significant ANOVA result (p = 0.0024), this **supports** the idea that **STEM preference was modestly linked to performance pre-program**.

ANOVA Results

- Pre-Assessment
  - F = 9.22, **p = 0.0024**
  - Significant: There is a **statistically significant** difference in scores between groups based on STEM interest.

05

Avg. Pre vs Post Assessment Scores by Favorite Subject

Chi-Squared Test Results: STEM Preference vs. Assessment Scores

- Post-Assessment
  - $X^2 = 5.72$, df = 10, **p = 0.838**
  - Interpretation:
    - This **p-value is very high**, indicating **no meaningful association** between STEM subject preference and score distribution after the program.
    - Low expected counts in some cells, so results should be **interpreted cautiously**.
    - However, this **aligns with the non-significant ANOVA result (p = 0.389)** and the score reversal observed between groups.

ANOVA Results

- Post-Assessment
- F = 0.744, **p = 0.389**
- Not significant: There is **no statistically significant difference** in post-assessment scores based on STEM interest.

05

Avg. Pre vs Post Assessment Scores by Favorite Subject

Data Completeness Caution

- The pre-assessment had only 1412 valid responses for this variable — 1784 were missing.
- The post-assessment had only 810 valid responses, with 1455 missing.
- While trends are interesting, **these insights should be interpreted with caution due to high levels of missing data**.

Pre-Assessment: Suggests a Small Link

- Students who **reported STEM** as their favorite subject **had higher average scores on the pre-assessment**.
- This difference was supported by a statistically significant ANOVA result and a borderline Chi-squared test, **suggesting a modest connection** between early STEM interest and academic performance.

Post-Assessment: No Relationship Found

- After the program, the difference in performance between students with and without STEM preferences disappeared entirely. In fact, **students who did not prefer STEM slightly outperformed** those who did, though the difference was **not statistically significant**.
- The Chi-squared test confirmed this, showing no relationship between STEM preference and score distribution ($p = 0.838$).

Program Impact May Have Equalized Performance

- This shift from a pre-program performance gap to post-program parity **suggests the intervention may have had an equalizing effect.**
- Rather than favoring those already inclined toward STEM, **the program appeared to lift learning outcomes across the board**, including for students less interested in STEM at the start.

05 Avg. Pre vs Post Assessment Scores by College

Figure 15:

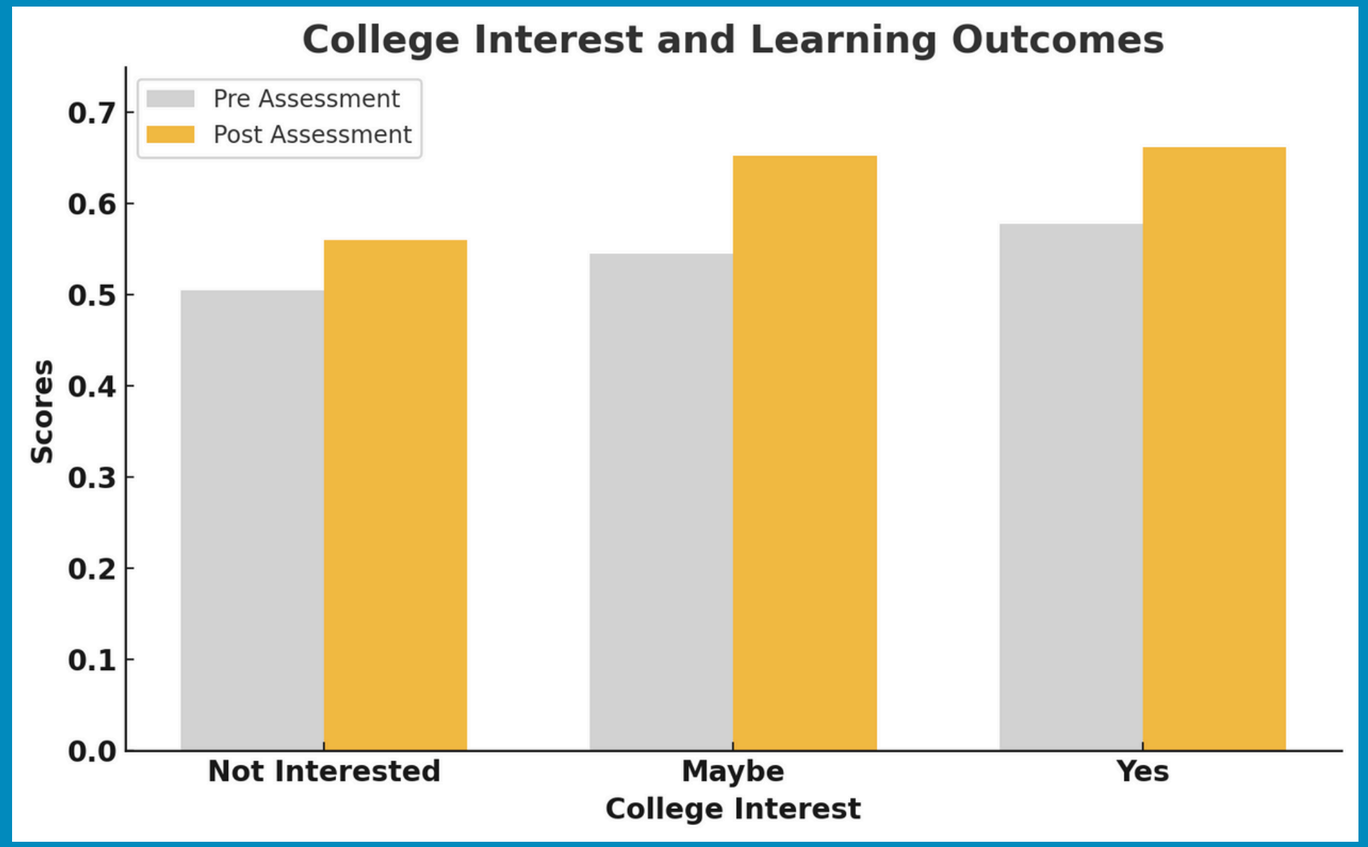


Figure 15 shows a bar chart comparing pre and post-mean scores based on students' interest in college.

College Interest

- **Students interested in college scored higher on both pre and post-assessments** compared to those unsure or not interested.
- **Improvement** was consistent across groups, but **college-interested** students had a slightly **higher gain in the 0.6–0.9 range.**

05 Avg. Pre vs Post Assessment Scores by College

Note on Missing Data in College Interest

- It is important to contextualize the observed difference in the number of responses to the college interest question between the pre-and post-assessments.
- In the pre-assessment, **only 59% of students answered the college interest** question (1892 of 3196). The remaining 41% were missing (NA).
  - As a result, any analysis involving college from the pre-assessment is based on partial data.
- In contrast, post-assessment response completeness improved dramatically: **91% of students answered the college interest** item (2053 of 2265).
  - Only 212 students (9%) had missing values in the post-assessment.
- **Why this matters:**
  - When comparing interest groups ("Not Interested", "Maybe", "Yes") over time, keep in mind that the pre-assessment groups may not fully represent the entire student population. This improvement reflects better survey completion and provides **more reliable insights in the post-assessment** regarding students' college interest and its relationship to learning outcomes.

05 Avg. Post Assessment Scores by College

- **College Interest Correlates with Higher Learning Outcomes**
  - Students who expressed interest in college had significantly higher average scores.
  - Even the **"Maybe"** group **outperformed** the **"Not Interested"** group by nearly **10 percent**.
  - The **"Yes"** group had the **highest** post-assessment **performance**, indicating a **positive relationship between college motivation and learning outcomes**.
- **Students with Lower College Interest Still Showed Gains**
  - Despite scoring lower on average, the **"Not Interested"** group **still improved** compared to their pre-assessment scores (from ~0.505 to 0.560).
  - This suggests the program impacts all students, not just those inclined toward academic success.
- **Post Assessment Data Is More Representative**
  - The post data reflects a broader, more representative sample of your student population.
  - As a result, post-assessment findings should be considered the stronger basis for concluding program effectiveness and the connection between motivation and performance.
  - It reveals a clear, positive relationship between students' interest in college and their learning gains, and **supports the conclusion that college motivation or program impact on college interest may play a key role in driving achievement.**

Table 11	
College Interest	Avg. Score
Not Interested (n=167)	0.56 (56%)
Maybe (n=677)	0.65 (65%)
Yes (n=1209)	0.66 (66%)

Table 11: Avg. Post Improvement Scores by College Interest Level

05 Avg. Post Assessment Scores by College

- **Pearson's Chi-squared Test**
  - The p-value is highly significant **($p < 0.001$)**, so we reject the null hypothesis.
  - This means **there is a statistically significant association between students' college interest levels and their assessment score performance.**
  - The distribution of scores differs meaningfully across college interest groups.
- **ANOVA (Analysis of Variance)**
  - The **F-value (18.04) is large, and the p-value (2.26e-05) is well below 0.001**.
  - This confirms that mean scores significantly differ by college interest level.
  - In other words, students' **motivation toward college is associated with higher or lower average performance** on the post-assessment.
- **Key Takeaways:**
  - Students who are more interested in college tend to score higher, on average.
  - The **pattern is not due to chance** and is likely to reflect a meaningful relationship between motivation and performance.

Figure 16:

```
        Pearson's Chi-squared test

data:  table
X-squared = 47.192, df = 20, p-value = 0.0005516

               Df Sum Sq Mean Sq F value   Pr(>F)
data$college    1   0.98  0.9781   18.04 2.26e-05 ***
Residuals    2042 110.73  0.0542
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
221 observations deleted due to missingness
```

05 Avg. Pre vs Post Assessment Scores by Coding

Figure 17:

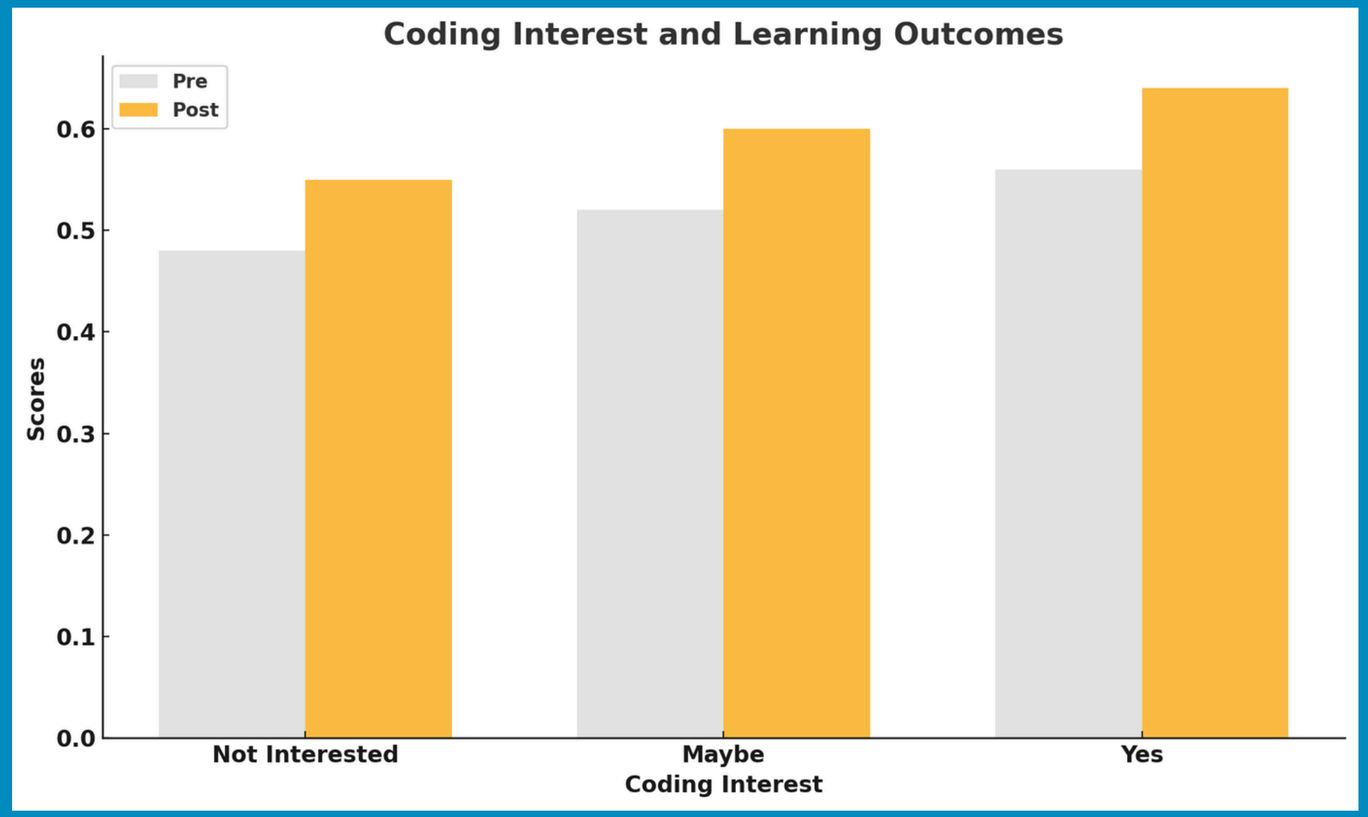


Figure 16 shows a bar chart comparing pre and post-mean scores based on students' interest in college.

Coding Interest:

- **Students interested in coding showed higher pre-scores** and maintained consistent improvement.
- Students **not interested** or **unsure** about coding exhibited **lower pre-scores** and **smaller improvements.**

05 Avg. Post Assessment Scores by Coding

- **Coding Interest Shows Small Differences in Learning Outcomes**
  - Students who expressed interest in coding had slightly higher average scores than those who were not interested.
  - However, the **differences between groups were small** — all three interest levels scored within a narrow range (0.618 to 0.649 on a 0–1 scale).
  - While the "Maybe" and "Yes" groups scored marginally higher, this was **not enough to indicate a strong link between coding interest and performance.**
- **Students Across All Coding Interest Levels Performed Similarly**
  - Despite differences in enthusiasm for coding, students across all interest levels **demonstrated comparable learning outcomes.**
  - This indicates that the program was **accessible and effective across varying levels of coding motivation.**
  - It's a positive sign that even students not especially drawn to coding can **still succeed in related content or learning experiences.**
- **Post Assessment Data Is More Representative**
  - The coding interest question was asked far more consistently in the post-assessment than in the pre-assessment. **Most students in the pre-assessment did not receive the coding interest question**, making that data largely incomplete.
  - In contrast, the post-assessment included this question for the majority of participants, resulting in 1328 valid responses (937 NA).
  - As such, analyses involving coding interest for **post-assessment data provide a more representative view of students' interest and their relation to learning outcomes.**

Table 12	
Coding Interest	Avg. Score
Not Interested (n=167)	0.62 (62%)
Maybe (n=677)	0.65 (65%)
Yes (n=1209)	0.65 (65%)

Table 12: Avg. Post Improvement Scores by Coding Interest Level

05 Avg. Post Assessment Scores by Coding

- **Pearson's Chi-squared Test**
  - A Chi-squared test of the distribution of scores across coding interest levels showed no meaningful association (p = 0.81).
- **ANOVA (Analysis of Variance)**
  - A one-way ANOVA revealed that the differences in mean scores by coding interest were not statistically significant (p = 0.214).
- **Key Takeaways:**
  - This suggests that students' performance on the assessment was not strongly influenced by their level of interest in coding.
  - Even though the "Maybe" and "Yes" groups have higher means, these differences could easily be due to chance.
- **Coding Interest May Reflect Preference, Not Performance**
  - Unlike college interest, which likely reflects academic or long-term motivation, **coding interest may not be tightly linked to assessment scores.**
  - It **could indicate a personal preference or career curiosity** that doesn't necessarily affect how well a student performs in a general assessment.

Figure 18:

```
        Pearson's Chi-squared test

data:  table
X-squared = 14.39, df = 20, p-value = 0.8102

               Df Sum Sq Mean Sq F value Pr(>F)
data$coding     1   0.08 0.08408   1.544  0.214
Residuals    1322  72.01 0.05447
941 observations deleted due to missingness
```

05 Avg. Scores vs College Interest by Program Type

Table 13										
College Interest	Virtual Reality		Circuits		HOM		Minecraft		CAD	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Not Interested	0.44	0.62	0.54	0.7	0.52	0.66	0.44	0.62	0.44	0.62
Maybe	0.53	0.56	0.54	0.56	0.54	0.56	0.54	0.56	0.52	0.58
Yes	0.63	0.7	0.63	0.7	0.67	0.7	0.68	0.7	0.77	0.74

Table 13: Analysis of Avg. Improvement Scores by College Interest Level After Disaggregating by Program Type

1. **Virtual Reality:**
   - Improvement: Students who were "Not Interested" showed the biggest jump, from 44% (pre) to 62% (post).
2. **Circuits:**
   - Improvement: Students who were "Not" showed the biggest jump, from 54% (pre) to 70% (post).
3. **Hands-on Modules:**
   - Improvement: Students who were "Not" showed the biggest jump, from 52% (pre) to 66% (post).
4. **Minecraft:**
   - Improvement: Students who were "Not" showed the biggest jump, from 44% (pre) to 62% (post).
5. **CAD:**
   - Improvement: Students who were "Not Interested" showed the biggest jump, from 44% (pre) to 62% (post).

- **Overall:** "Not Interested" students improved at higher rates than any of the three categories. All programs had a higher avg. score for all levels of interest, except CAD, whose "Yes" students avg. decreased to 74%

05 Avg. Scores vs Coding Interest by Program Type

Table 14										
Coding Interest	Virtual Reality		Circuits		HOM		Minecraft		CAD	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Not Interested	0.56	0.6	0.5	0.56	0.62	0.64	0.57	0.6	0.62	0.64
Maybe	0.6	0.62	0.64	0.7	0.64	0.7	0.63	0.67	0.64	0.7
Yes	0.48	0.67	0.48	0.67	0.55	0.68	0.55	0.68	0.55	0.68

Table 14: Analysis of Avg. Improvement Scores by Coding Interest Level After Disaggregating by Program Type

1. **Virtual Reality:**
   - Improvement: The largest improvement was seen in "Yes Interested" students, jumping from 48% (pre) to 67% (post).
2. **Circuits:**
   - Improvement: "Yes" students improved the most, especially in post-assessment scores (67%).
3. **Hands-on Modules:**
   - Improvement: This program showed moderate gains across all groups, particularly for "Yes Interested" students. “Maybe” did best at 70%.
4. **Minecraft:**
   - Improvement: The "Yes" group saw the greatest improvement, moving from 55% (pre) to 68% (post).
5. **CAD:**
   - Improvement: "Yes" students in CAD programs showed steady improvement, with post-assessment scores reaching 68%. “Maybe” did best at 70%

06 Suggestions

1. Standardized Survey Design

- **Uniform Question Wording, Structure, and Scaling:** Ensure that core questions (e.g., college interest, coding interest, learning confidence) remain the same year over year, avoiding extra spaces or word changes to allow direct comparisons. Have a three-point scale vs a 5 or 2-point scale (e.g. Yes, Maybe, No).
- **Consistent Survey Sections:** Keep the same section structure across pre- and post-surveys (e.g., demographics, engagement, learning outcomes, program impact).
- **Pre vs. Post Differentiation:** Include a clear question or indicator to distinguish pre-assessment from post-assessment to avoid misclassification in analysis.
- **Track Courses More Accurately:** Ensure students select only courses they participated in (e.g. VR, math, Minecraft, etc.) (instructors can guide them before they complete the survey).
- **Grade & College/Coding Interest Always Included:** Ensure grade level and college/coding interest are asked in every survey (some past versions omitted them).

2. Enhancing Data Accuracy & Completeness

- **Mandatory Core Questions:** Require key fields (e.g., grade level, program type, interest changes) to reduce missing data.
- **Reduce Survey Variations by Year:** Minimize unnecessary wording changes in questions that could impact longitudinal analysis.
- **Add a Survey Progress Indicator and Reply Limit:** This helps students understand how much is left to complete, reducing dropout rates before submission. Set a one-reply limit. Having unique IDs would help.

06 Suggestions

3. Improved Student Tracking While Maintaining Anonymity

- **Unique Student IDs:** Assign randomized IDs at the start of the program that students write down and use for both pre-and post-surveys.
- **Reduce Open-Ended Responses:** Convert subjective, open-ended responses into structured "select all that apply" options where possible (e.g., favorite subject).
- **Ensure Every Survey Captures Core Demographics:** Collect grade level, language preferences, and previous STEM program participation in every survey, not just some.
- **Cohort & Program Identification:** Include session codes (e.g., Summer 2024, Winter 2025) to track results by group.

4. Analyzing Data for Actionable Insights

- **Break Down Learning Gains by Subgroups:** Analyze trends by grade level, gender, and prior experience in STEM programs.
- **Continue Assessing Course Effectiveness:** Compare pre- and post-scores per course (VR, Circuits, CAD) to identify strengths and weaknesses in the curriculum.
- **Engagement Metrics Standardization:** Keep "likelihood of recommending" and "enjoyment of program" phrased consistently for trend analysis.

06 Suggestions

5. Variables to Emphasis and/or Add

- Level of Engagement (1-3 scale)
- Instructor Feedback
- College Interest
- Coding Interest
- STEM interest
- Program Recommendation
- Program Renewal
- Program Enjoyment
- Past Program Participation
- Cohort Group (e.g. Ventura, CHE, etc.)
- Randomized Unique Temporary IDs (this will remove the need for names and help with matching pre and post-surveys. Having a list of random numbers already generated would help avoid repeats as well)
- Cohort Season
- Pre vs Post Assessment
- Gender
- Languages Spoken
- Languages Preferred
- Favorite Subject (select all that apply)

THANK YOU